Log Book Entry 5.2

NB2420 Electronic Instrumentation · Manual 5.2 · Differential measurements

Hand in via BrightSpace

Name(s): | Partner:

Date: | Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

BALANCE AND COMMON-MODE REJECTION

Tasks I1, I2, I3 and C1

Keep predictions and simulation traces beside this entry. Record raw readings before calculating gains. Voltages are in V.

Subtractor input test Task I2, reference at ground

Nominal input pair	U_L measured	U_R measured	U_{out} measured
0.5, 1.5			
1.5, 0.5			
0.5, 0.5			
1.5, 1.5			

Subtractor quantity	Predicted, matched	Simulated, matched	Measured
A_d			
A_c			
CMRR (dB), or bound			

Measurement resolution used for the common-mode result:

Equal illumination Tasks I1 and C1

Light level	U_L before	U_R before	U_L after	U_R after
Lower				
Higher				

Left feedback before / after (kΩ):

Right feedback (kΩ):

Imbalance follows sensor or channel after swap?

Full detector Task I3

Torch condition	U_L	U_R	U_{out}
Left			
Centre			
Right			

Loaded U_{ref} (V):

Verified output minimum / maximum (V):

☐ ADC2 connected only after checking the output range and common ground.

Setup sketch, trace or display file reference; include any deep dive evidence:

FIVE-LEVEL DETECTOR

Tasks I4 and C2

Baseline (software units):

Largest calibration departure (software units):

Deadband D / strong threshold H (software units):

Use one baseline for the comparison. Record a limitation if a level cannot be reached within the verified ADC range.

Torch condition	U_{out} (V)	Corrected reading	Displayed level and label
Far left			
Left			
Centre			
Right			
Far right			

Shared-light check after trimming No directional torch; keep the same baseline

Shared illumination	U_{out} (V)	Corrected reading	Displayed level
Lower			
Higher			

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: